WORKED SOLUTION

STEP II 1996, Question 12

Probability and Statistics • Maxima and Minima Distributions

STEP II 1996, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q12

The key idea

The maximum of several independent variables is awkward to handle directly, but the event "the maximum is small" is easy: it happens exactly when *every* variable is small, and independence lets us multiply. So we find the cumulative distribution function first and differentiate to get the density. Part (ii) is the mirror image: the *minimum* is large exactly when every variable is large.

Solution

Part (i)

Each X_j is uniform on $[0,1]$, so $\mathrm{P}(X_j < y) = y$ for $0 \le y \le 1$. For y in this range, $Y < y$ happens if and only if $X_j < y$ for every j, and since the X_j are independent,

$$F(y) = \mathrm{P}(Y < y) = \prod_{j=1}^{n} \mathrm{P}(X_j < y) = y^n, \qquad 0 \le y \le 1.$$

Of course $F(y) = 0$ for $y < 0$ and $F(y) = 1$ for $y > 1$. Differentiating, the probability density function of Y is

$$f(y) = \begin{cases} ny^{n-1}, & 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

(As a sanity check, $\int_0^1 ny^{n-1}\,\mathrm{d}y = 1$.)

Now the moments:

$$\mathrm{E}(Y) = \int_0^1 y \cdot ny^{n-1}\,\mathrm{d}y = \frac{n}{n+1},$$
$$\mathrm{E}(Y^2) = \int_0^1 y^2 \cdot ny^{n-1}\,\mathrm{d}y = \frac{n}{n+2}.$$

Hence

$$\mathrm{Var}(Y) = \frac{n}{n+2} - \frac{n^2}{(n+1)^2} = \frac{n\left[(n+1)^2 - n(n+2)\right]}{(n+2)(n+1)^2} = \frac{n}{(n+2)(n+1)^2},$$

since $(n+1)^2 - n(n+2) = n^2 + 2n + 1 - n^2 - 2n = 1$. This is the required result.

Ask yourself Why work with $\mathrm{P}(Y < y)$ rather than trying to write down $\mathrm{P}(Y = y)$ or a density for Y straight away?

For a continuous variable the density isn't a probability, and "the largest equals y" mixes up which X_j is largest with what the others do. The event $\{Y < y\}$ is an intersection of n independent events, so its probability is a simple product. Getting the distribution function and then differentiating is the standard route for maxima and minima.

Part (ii)

Let T_j be the failure time of the jth light, so $\mathrm{P}(T_j \le t) = 1 - \mathrm{e}^{-t/\lambda}$ and

$$\mathrm{P}(T_j > t) = \mathrm{e}^{-t/\lambda}, \qquad t > 0.$$

The lights are independent, so the T_j are independent. Let $T = \min(T_1, \ldots, T_n)$ be the time of the first failure. This time the convenient event is $T > t$, which happens if and only if every light is still working at time t:

$$\mathrm{P}(T > t) = \prod_{j=1}^{n} \mathrm{P}(T_j > t) = \left(\mathrm{e}^{-t/\lambda}\right)^n = \mathrm{e}^{-nt/\lambda}.$$

So $\mathrm{P}(T \le t) = 1 - \mathrm{e}^{-nt/\lambda}$, and differentiating gives the density

$$g(t) = \frac{n}{\lambda}\,\mathrm{e}^{-nt/\lambda}, \qquad t > 0.$$

In other words, T has the same kind of distribution as a single light, with λ replaced by λ/n. Integrating by parts with $u = t$,

$$\begin{aligned}\mathrm{E}(T) &= \int_0^\infty t\,\frac{n}{\lambda}\,\mathrm{e}^{-nt/\lambda}\,\mathrm{d}t = \Big[-t\,\mathrm{e}^{-nt/\lambda}\Big]_0^\infty + \int_0^\infty \mathrm{e}^{-nt/\lambda}\,\mathrm{d}t \\ &= 0 + \Big[-\frac{\lambda}{n}\,\mathrm{e}^{-nt/\lambda}\Big]_0^\infty = \frac{\lambda}{n},\end{aligned}$$

where the boundary term vanishes because $t\,\mathrm{e}^{-nt/\lambda} \to 0$ as $t \to \infty$. So the expected time until the first failure is λ/n, as required.

Checks

- With $n = 1$, Y is just uniform on $[0,1]$, and the formula gives $\frac{1}{3\cdot 4} = \frac{1}{12}$, the familiar variance of a uniform variable.

- As n grows, the variance behaves like $1/n^2$: the maximum of many uniforms is squeezed up against 1, as you'd expect.
- With $n = 1$ in part (ii), the expected lifetime of one light is λ, which is the mean of an exponential distribution with parameter $1/\lambda$.
- A simulation of 200 000 maxima of five uniforms gave variance 0.01993, against $\frac{5}{7 \cdot 36} \approx 0.01984$; simulating four lights with $\lambda = 3$ gave a mean first failure time of 0.749, against $\frac{3}{4}$.

Common slips

- Differentiating $\mathrm{P}(T > t)$ and forgetting the minus sign, which gives a negative "density".
- Writing $\mathrm{Var}(Y) = \mathrm{E}(Y^2) - \mathrm{E}(Y)$ instead of subtracting $(\mathrm{E}(Y))^2$.
- Thinking that n lights should fail on average at time λ each, so the first fails at λ. The minimum of n variables is much smaller than any one of them.

Beyond STEP

The maximum and minimum are the extreme cases of *order statistics*: if you sort a sample, the kth smallest of n uniforms has a beta distribution, and Y here is the case $k = n$. Part (ii) shows the key fact that the minimum of independent exponential variables is again exponential, with the rates adding. This drives the theory of Poisson processes and continuous-time Markov chains. Grimmett and Welsh, *Probability: An Introduction*, and Blitzstein and Hwang, *Introduction to Probability*, both develop these ideas.

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